

If you are studying physics in high school then refer *Physics 5th* edition by Halliday, Resnick, Krane. This guide is intended for those who want to complete a undergraduate degree in physics (I will be focusing on theoretical and mathematical physics). I am assuming you have mathematical knowledge up until mentioned in this [PDF](#). We start with introductory physics and then work our way through the undergraduate core as well as electives you may take.

1. Mathematics that you will encounter throughout your undergraduate degree.
 - *Mathematical Methods in the Physical Sciences* by Boas
 - *Mathematical Methods for Physicists* by Arfken, Weber, Harris
2. We go through basic mechanics, electrostatics and magnetostatics and modern physics.
 - *Introduction to Mechanics* by Kleppner, Kolenkow and *Introduction to Classical Mechanics* by Morin
 - *Electricity and Magnetism* by Purcell, Morin [supplement with *Div, Grad, Curl, and All That* by Schey]
 - *Vibrations and Waves* by French and *Waves* by Crawford
 - *Introduction to Quantum Mechanics* by Griffiths, Schroeter and *A Modern Approach to Quantum Mechanics* by Townsend
3. We go through the undergraduate core of a physics degree.
 - *Classical Mechanics* by Taylor [supplement with *Notes on Dynamics* by Tong]
 - *Intorduction to Electrodynamics* by Griffiths [supplement with *Notes on Vector Calculus* by Tong]
 - *Thermal Physics* by Schroeder or *Fundamentals of Statistical and Thermal Physics* by Reif
 - *Optics* by Hecht
 - *Principles of Quantum Mechanics* by Shankar
4. We can now attempt undergraduate electives in physics which you can check out at this [post](#).

One can only learn physics by solving problems so here is list of problem books to go through throughout your undergraduate degree.

- *Problems in General Physics* by Irodov
- *200 Puzzling Physics Problems* by Gnädig, Honyek, Riley
- *Physics Olympiad: Basic to Advanced Exercises* by The Committee of Japan Physics Olympiad
- *Princeton Problems in Physics with Solutions* by Newbury
- *A Guide to Physics Problems, Part 1* by Cahn, Nadgorny
- *A Guide to Physics Problems, Part 2* by Cahn, Nadgorny
- *Problems and Solutions on Mechanics* by Lim Yung-Kuo
- *Analytical Dynamics of Particles and Rigid Bodies* by Whittaker
- *Problems and Solutions on Electromagnetism* by Lim Yung-Kuo
- *Static and Dynamic Electricity* by Smythe
- *Problems and Solutions on Quantum Mechanics* by Lim Yung-Kuo
- *Problems and Solutions on Thermodynamics and Statistical Mechanics* by Lim Yung-Kuo
- *Problems and Solutions on Optics* by Lim Yung-Kuo
- *Problems in Quantum Mechanics with Solutions* by Squires
- *Problems in Quantum Mechanics* by Constantinescu, Magyari
- *Exploring Quantum Mechanics* by Galitski, Karnakov, Kogan
- *Problems in Electrodynamics* by Batygin, Toptygin
- *Statistical Mechanics: An Advanced Course with Problems and Solutions* by Kubo
- *Schaum's Outline of Optics*